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Bengali Text Categorization Based on Deep Hybrid CNN–LSTM Network with Word Embedding

Samrat Alam
Department of Computer Science & Engineering
Khulna University of Engineering and Technology
Khulna-9203, Bangladesh
samratalamshanto710@gmail.com

Md Afnan Ul Haque
Department of Computer Science & Engineering
Khulna University of Engineering and Technology
Khulna-9203, Bangladesh
nuhash97afnan@gmail.com

Ashiqur Rahman
Department of Computer Science & Engineering
Khulna University of Engineering and Technology
Khulna-9203, Bangladesh
rahmanashiqur581@gmail.com

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***Abstract*— With the evolution of the internet and digital network, the amount of textual data is increasing gradually. As a result, it becomes increasingly difficult to categorize vast amounts of data manually. But, with the benefit of the machine learning process, we can categorize these massive amounts of text data automatically according to their contents. Compared with other languages such as English, text categorization in Bangla is one of the challenging tasks. Because a lack of resources is the reason for this. In this research work, we take data from open resources which are containing the Bangla newspaper article corpus with 12 categories. For categorizing these article datasets, we have used word embedding methods to extract features from raw data and then used deep learning methods: Convolutional Neural Network (CNN), Artificial Neural Network (ANN), Long Short-term memory (LSTM) and Hybrid (CNN+Bi-LSTM) methods to classify them based on their contents. Next, we have analyzed the performance of these deep learning based classification models and looked into the reasons for it. Furthermore, the hybrid model has a greater accuracy of 88.56% in 10 categories and 84.93% in 12 categories. Lastly, the limitations of this study as well as future prospects have been discussed.**

Keywords*— *Text Classification, Bangla, Deep Learning, Word Embedding, Hybrid model.

I. Introduction

By taking the advantage of the evolution of the Internet and modern technology, digital data is growing at high speed over time [1]. According to International Data Corporation (IDC), it is estimated that by 2025, the total amount of the world's data will be 175 zetta-bytes (ZB) and the growth rate will be around 61 percent. In most cases, the vast amounts of digital text data are uncategorized and contain a wide range of irrelevant material. As a result, the exponential growth of these uncategorized text data is a serious problem for the recent era [2]. Therefore, the need for text data categorization is an essential and demanding topic in natural language processing (NLP).

Text Categorization is the process of categorizing text data into different classes depending on their contents [3]. The main goal of this process is to extract the features of the text and categorize them based on features. There are a lot of machine learning algorithms, such as Logistic Regression (LR), Naive Bayes, Support Vector Machines (SVM), etc. as well as deep learning approaches such as RNN, CNN, LSTM based model, etc. for this text classification. It has been proved that deep learning methods can extract useful features from raw text data among all methods [4]. That is why they carry out better performance in this case.

Bangla is the sixth most widely spoken language in the world. Approximately 265 million people all over the world speak this language [5]. According to the annual report of the Bangladesh Telecommunication Regulatory Commission (BTRC), there was around 96.199 million internet user of this Bangla language at the end of July 2019 [6] and the number of users, as well as newspaper readers, are steadily expanding. As a result, there are huge Bengali data in social media, newspapers, web pages about art, economics, history, opinions, politics, sports, religion, and so on. However, compared to other languages such as English, Bengali has fewer resources for text classification. There is a shortage of quality datasets in Bengali with proper labels. Some datasets are not standard level. However, the Bengali language contains 24 types of forms for pronouns, 36 different noun types and 160 types of inflected verb forms [7]. So, it is clear that Bengali text categorization is not an easy task and becomes a big challenge.

In this paper, we have used an open-source dataset for this Bangla text categorization. The data is collected from Bangla newspaper articles and there are 12 categories. The twelve categories are 'Accident', 'Art', 'Crime', 'Economics', 'Education', 'Entertainment', 'Environment', 'International', 'Opinion', 'Politics', 'Science & Technology' and 'Sports'. After some processing like stemming, removing stop words, punctuation on this text data, we have used them to train some deep learning models. In recent times, deep learning models are gaining very much popularity for achieving tremendous performance. It learns important features from the data and then processes them according to the tasks. In this paper, we have provided an overview of some widely used deep learning models: CNN, LSTM, ANN for the Bangla text dataset. We have also merged CNN and Bi-LSTM models and trained with the dataset. Then we have evaluated the performance among all these deep learning based classification models.

The rest of the paper is organized as follows: Related works have been elaborated in section II. The methodology with 5 sub-sections is described shortly in section III. Section IV presents the Experimental Analysis with 4 sub-sections.

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Finally, the conclusion with future aspects has been drawn in section V.

II. Related Work

There are a lot of works in the field of automated text classification in recent times. Especially, in the English language, a great amount of study in this text classification had done because of vast resources in text processing. However, there are insufficient resources for text processing in the Bangla language. As a result, Bangla text classification has proven to be a difficult task. Let us look at a few works and discuss about them:

Tudu et al. [8] used a public dataset of Bengali newspaper articles that contains 84,906 Bangla text data in 10 categories. In this paper, they used some machine learning approaches: Support Vector Machine (SVM), Multinomial Naive Bayes (MNB), Stochastic Gradient Descent (SGD) and Logistic Regression (LR) to classify the Bangla text data. First, they divided the dataset into 10027, 42370, 60000 and 84906 and then classified them. They found that the accuracy decreased for increasing the number of training data. They got the highest accuracy of 87.5% from SVM for 84906 sample data.

In this research paper [9], Chowdhury et al. addressed a methodology for news data classification based on the news headlines. They collected the dataset from Kaggle. In this dataset, there were around 14k news data into 10 categories. They used 'GloVe' for vectorization to achieve better accuracy. They applied a hybrid deep learning model: LSTM and CNN for feature extraction. They got the highest 87% accuracy using 'GloVe' and 79% accuracy without 'GloVe' from their proposed method on the Kaggle news dataset.

The authors of [10] used the same open source dataset of Tudu et al. [8] had used. First, they selected 250 articles from each category to make a balanced dataset. They did not use the full dataset. They took only 3000 articles from 95853 articles. They used Support Vector Machine (SVM), Random Forrest, Logistic Regression, BiLSTM and CNN. They got higher accuracy from the CNN model for their small balanced dataset.

Chakraborty et al. [11] collected 1600 articles from two online newspapers and categorize them into 16 categories. Each category had a total of 100 amounts of articles. Then used TF-IDF to extract the features from text data. They proposed a model with three dense layers and they used TF-IDF features as an input for the first dense layer. They used some other models (machine learning classifier and neural networks) to compare with the proposed model. They found that their proposed model performed well than other models. They got the highest accuracy of 84.58% from the proposed model.

In this study, we have used the same dataset as [8], [10] used in their works. Then we have applied some deep learning approaches to this large Bangla dataset and categorized them based on its contents. We also have used a hybrid model which is a combination of the CNN and BiLSTM models. Finally, the results were compared to each other, as well as to other research studies.

III. The Methodology

In this section, we have discussed the working procedure of this work. This section is divided into 5 sub-sections. The complete working procedure is shown in the following Fig. 1.

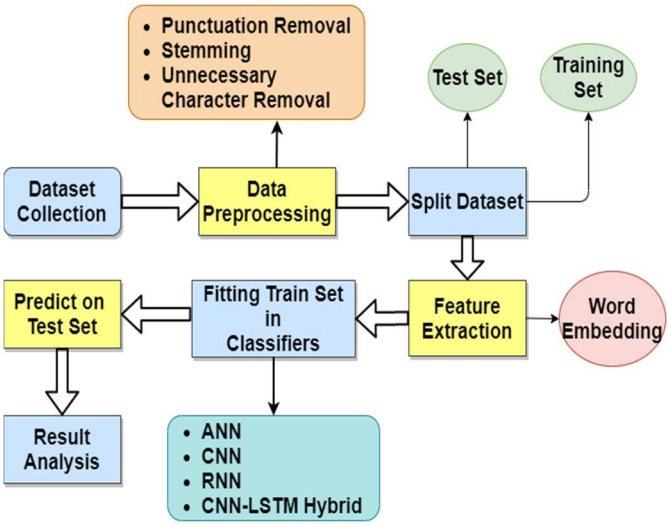

Fig. 1. The working procedure of the methodology.

A. Dataset:

We have selected an open-source public dataset that was used in this research work [8]. Mainly, this dataset is created from Bangla newspaper articles. Dataset can be accessed at https://corpus.scdnlab.com/ and can be found online as titled বাংলা ডাটাসেট. This dataset is a thesis work under the Department of Computer Science and Engineering in Shahjalal University of Science & Technology (SUST), Bangladesh. This dataset is published under the MIT license. So, anyone can use this data for research purposes. There are a total of 95853 text data in 12 categories. The categories are Accident, Art, Crime, Economics, Education, Entertainment, Environment, International, Opinion, Politics, Science & Technology and Sports.

B. Data Preprocessing:

At first, we want to remove noise and irrelevant things from the data. As the data is Bengali text, the text usually carries varied types of noises. This noise creates a problem in the process of feature engineering so we need to remove it from the text and preserve the main information as much as possible. We have taken some rudimental steps to remove noise. They are:

1) Stopwords Removal:

Stop words are known as the list of most common words in a language. They are typically removed from the texts as they do not have very much contextual meaning or they contribute towards any feature extraction techniques. In the Bengali language, there also exist stop words. But as a low resource language, there are no benchmark stop words collection list. So we created a stop words list consisting of 732 words. They are selected manually and checked before use. We will release it publicly for further research purposes. Some of them are given below:

অথবা, অন্য, আর, ইত্যাদি, এই, এছাড়াও, কিছু, জন, তথা, তারপর, সেই, সুতরাং, যদি

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2) Punctuation Removal:

We have removed all the punctuations in the text as they don't contribute anything to the feature extraction process. There exist different types of punctuation that are used in the Bengali language. In fact, there are different Unicode values for the same punctuation used. So we have used regex to remove these punctuations.

3) Unnecessary Character Removal:

There are some characters in the text data that are not punctuations but they are unnecessary. As an example ('<', '>', '@', '#', '$', '%', '^', '&', '*', '+', '\', '/'). We have removed these characters from our text dataset.

4) Stemming:

Stemming is an important part of natural language processing. Stemming means converting a word to its root word form. It reduces the dimensionality of the data. As Bengali, a low-resource language, there exist very few good stemmers. So we used the 'bangla-stemmer 1.0' in this paper. As it is not a very good stemmer, so we used it multiple times in a sentence that works better. An example of the Bangla stemming process is illustrated in fig. 2.

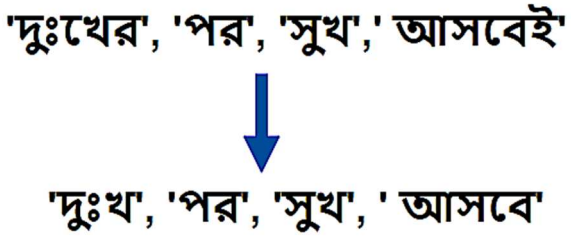

Fig. 2. An example of the bangla stemming process.

5) Emotion Removal:

In social media platforms, it is very common to practice nowadays to use emoticons in text. They have become a common thing in the text. Though they have some sentiment or meaning to offer they can be used for the useful feature. But as Bengali is a low-resource language, there is no library available to use an emoticon. So we had to remove it from the dataset using regex.

C. Split Dataset:

After the preprocessing is done, we split the data into an 80-20 ratio in a stratified fashion. By using a stratified technique, we have maintained the percentage of data points for each label in both the train and test set.

D. Feature Extraction:

1) Word Embedding:

Word embedding is nowadays a vital technique of natural language processing that plots a word's semantic meaning into a vectorized geometric space. After tokenization of unique words, vectors are created from the distance between two correlated words based on their semantic meaning. A neural network model is associated to create word embedding. Word embedding is nothing but n-dimensional vectors that contain a word's semantic meaning in a geometric space. For example, 'মা' (mother) and 'বাবা' (father) words have a similar semantic meaning and their corresponding word embedding vectors will be similar to each other. But 'পাখী' (Bird) word is very different from 'মা' (mother) and 'বাবা' (father). As similar words have the same vectors, learning models can easily get the context of words from the n-dimensional vectors. Let us see an example of the word embedding process. In fig. 3, the word embedding process is shown with an example.

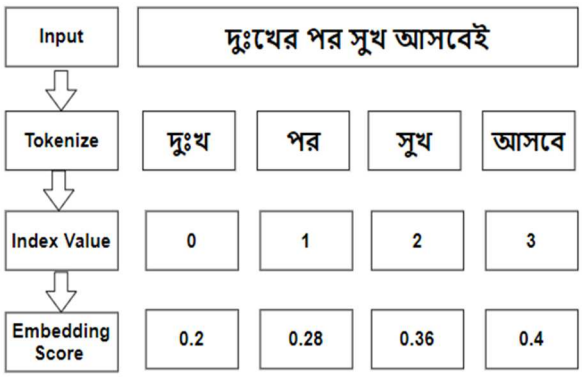

Fig. 3. The word embedding process.

E. Classification Models:

1) ANN:

The neuron is known as the mathematical unit that computes some linear function from the input sequence it takes and produces some output sequence that is passed to the next neuron. A neural network is a set of layers (layer consists of some neurons) placed side by side in a consecutive manner. The neuron of a layer is connected by an edge to all neurons in the previous and next layers. Neurons and their edges have some weights that reconcile by maximizing or minimizing it with the learning yield. In our work, at first, we have tokenized the texts. After that, we have converted the texts into vectors where each word is replaced with its token value. Vectors with token values are passed into embedding layers to find the features within these texts. After a flatten layer, we feed these vectors into hidden layers. These layers find the features in the text vectors. The final layer is used with softmax activation, shows the highest value that is our predicted class by the neural network.

2) RNN:

Recurrent neural networks (RNN) are a type of network which is a class of neural networks where connected nodes create a directed graph on a temporal sequence. It is a state-of-the-art algorithm that has internal memory and it can remember the sequence. It takes a decision combining the current input and the previous input. So it is more robust. In-text categorization problems in machine learning, data points are considered long and have context in their length. So it needs to be remembered by the algorithm. RNN is perfect for text categorization problems as it has internal memory and remembers long text sequences. RNN has a different type of variant such as GRU (Gated Recurrent Unit), LSTM (Long Short-Term Memory). LSTM architecture can be both unidirectional and bi-directional in nature. In Unidirectional LSTM, the input sequence is passed from backward to forward and it only preserves the context of the past. But in bi-directional LSTM, the input sequence runs from backward to forward and then forward to backward through the two hidden states combined. This helps LSTM to remember the sequence context not only from the past but also from the future. In Bidirectional LSTM, it is normally a RNN that is

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configured in a bidirectional way. The two ways are backward and forward directions, they are combined with any preserved information from the past and future. In our work, we have used Bi-LSTM architecture as an RNN unit. The basic LSTM block is shown in fig. 4. where X_t is the input vector, h_{t-1} is the previous cell output, C_{t-1} is the previous cell memory, h_t is the current cell output and C_t is the current cell memory.

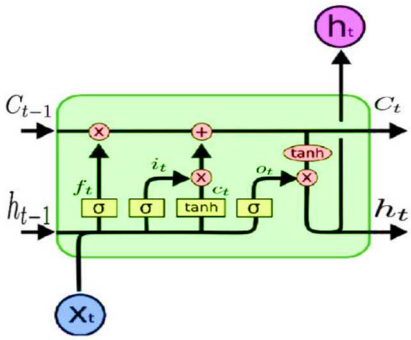

Fig. 4. The basic structure of the lstm block.

3) CNN:

Convolutional Neural Network (CNN) is one of the widely used deep learning models for extracting the best features from text data. There are several layers like the Convo1D layer, MaxPooling1D layer, Flatten layer and Dense layer in the architecture of the CNN model. In CNN, the embedding layer is the input layer. Then the Convo1D layer and Maxpooling1D layer are used to extract local features from the data and reduce the complexity. The next layer is the Flatten layer which converts the two-dimensional features into vector form for the next layer: a dense layer. The dense layer is mainly a fully connected layer with an activation function. ReLU and Sigmoid are two activation functions used in the dense layers of this CNN model. The last dense layer with the sigmoid activation function performs the classification task.

4) CNN-LSTM Hybrid Architecture:

Sometimes in NLP applications, hybrid networks have proven good results [12]. In large text data, CNN can extract the local features from the vector representation from the word embedding of the text and combine these local features. Bi-LSTM can extract global features based on those local features and memorize them. So, we have combined CNN and Bi-LSTM as it is a promising architecture and proved good results. The model summary of this hybrid network model with input and output shapes is provided in fig. 5.

IV. Experimental Analysis

In this section, we have discussed the results of the deep learning models for the Bangla text classification dataset. We have divided this section into four parts, dataset description, model and parameters, result discussion and performance evaluation of the deep learning models.

A. Dataset Description:

There are 95853 text data in 12 categories in this data set. All of the data was gathered from newspapers. The number of data per class has illustrated in Table I. And Table I shows that the dataset has the most data in the 'Politics' class and the least data in the 'Art' class.

TABLE I: NUMBER OF TEXT DATA PER CLASS IN THE DATASET

Class	Number of text data
Accident	6126
Art	2170
Crime	8199
Economics	3389
Education	10706
Entertainment	8951
Environment	3754
International	5036
Opinion	7684
Politics	18558
Science & Technology	2810
Sports	11296

B. Model and Parameters:

In this study, we have tuned parameters for increasing the accuracy rate of the model. In Table II, we have displayed the model summary and the best parameters of this study.

TABLE II: THE PARAMETERS OF DEEP LEARNING MODELS

Parameters	Best
ANN: Embedding= [300,400,50], Layers= [7,8,9,12], Activation Function= [softmax,relu], Learning rate= [0.005,0.0005,0.00005], Batch size= [64,128,256], Regularizer= [L2], Dropout= [0.5,0.2], Optimizer= [RMSprop,Adam]	Embedding= [400], Layers= [9], Activation Function= [relu], Learning rate= [0.0005], Batch size= [128], Regularizer=[L2], Dropout= [0.5], Optimizer= [Adam]
CNN: Embedding= [300,400,500], Conv1d= [100,200,300], Maxpooling1d= [2], Learning rate= [0.005,0.0005,0.00005], Batch size= [64,128,256], Regularizer= [L2], Optimizer= [RMSprop,Adam]	Embedding= [400], Conv1d= [200], Maxpooling1d= [2], Learning rate= [0.0005], Batch size= [128], Regularizer= [L2], Optimizer= [Adam]
LSTM: Embedding= [300,400,500], Bi-LSTM Layers= [2,3], Bi-LSTM Units= [64,128], Learning rate= [0.005,0.0005,0.00005], Batch size= [64,128,256], Regularizer= [L2], Optimizer= [RMSprop,Adam]	Embedding= [400], Bi-LSTM Layers= [2], Bi-LSTM Units= [64], Learning rate= [0.0005], Batch size= [128], Regularizer= [L2], Optimizer= [Adam]
LSTM-CNN Hybrid: Embedding= [300,400,500], Conv1d= [100,200,300], Maxpooling1d= [2], Bi-LSTM Layers= [2,3], Bi-LSTM Units= [64,128], Learning rate= [0.005,0.0005,0.00005], Batch size= [64,128,256], Regularizer= [L2], Optimizer= [RMSprop,Adam]	Embedding= [400] Conv1d= [200], Maxpooling1d= [2], Bi-LSTM Layers= [2], Bi-LSTM Units= [64], Learning rate= [0.0005], Batch size= [128], Regularizer= [L2], Optimizer= [Adam]

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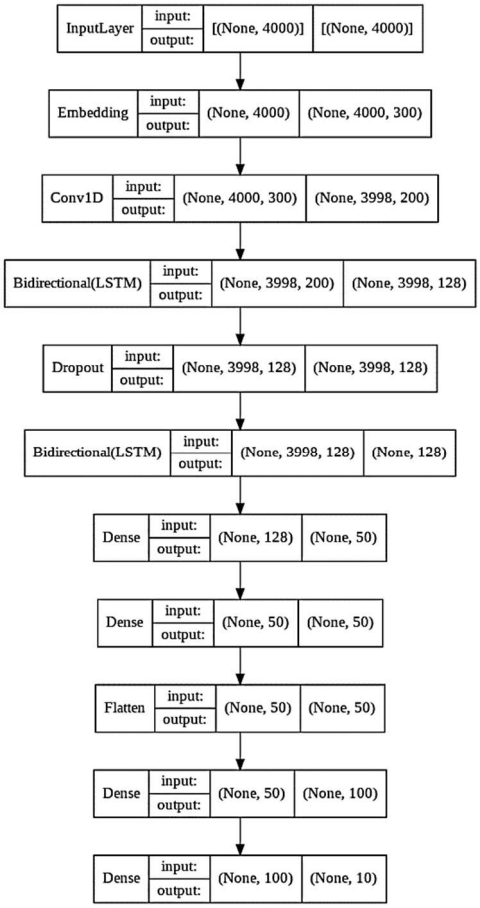

Fig. 5. The model summary of the hybrid network model.

C. *Result Discussion:*

We have divided this sub-section into two parts, one is for comparing results among the discussed models and another one compares with other results on this Bangla text classification dataset.

a) Results Analysis among discussed Models:

In this research paper, we have discussed some deep learning models. First, we have divided the data into an 80-20 ratio in a stratified fashion. Then we have trained these models with the training dataset. In fig. 6, the accuracy of these deep learning models for 12 text categories has been demonstrated.

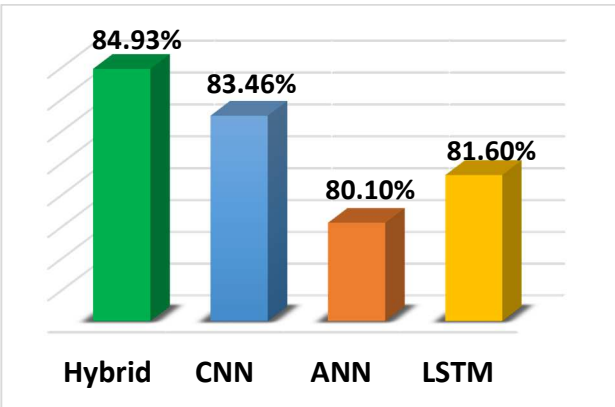

Fig. 6. The accuracy between different deep learning models.

Here, we have got higher accuracy (84.93%) from the hybrid model for this 12 categories Bangla text dataset and the lowest (80.10%) from ANN based model. The hybrid model is the combination of these CNN and Bi-LSTM models. Mainly, the CNN model extracted the significant features from the given training text data and then the Bi-LSTM layer extracted content based features and learned them. This model works best for this dataset. CNN model is worked best for small datasets for extracting local features perfectly. That is why we have got higher accuracy from CNN than the LSTM based model for this small dataset. When the dataset is large, CNN cannot memorize the long term dependencies. But LSTM can do this and achieve good results for large datasets. This dataset contains the same text data into different categories. As a result, when dealing with this type of text input, models are more prone to misclassification. Furthermore, the Bangla language lacks an effective stemmer. Therefore, it is impossible to eliminate redundant features by reducing the different forms of Bangla word roots. As a result, these models are unable to achieve greater accuracy for these 12 categories Bangla dataset.

b) Results Analysis with other Results:

In this sub-section, we have made a comparison with Tudu et al. [8]. They worked with 10 categories. So, we have used the same dataset. First, we have removed the Art and Opinion category from the dataset like them [8]. Then we have compared with Tudu et al. [8]. In Fig. 7, the best method accuracy comparison for these 10 categories of this dataset is depicted. For this 10 categories dataset, our hybrid model provided the maximum accuracy. They applied some machine learning classifiers: SVM (Support Vector Machine), LR (Logistic Regression), SGD (Stochastic Gradient Descent) and MNB (Multinomial Naive Bayes) and they got the best result from the SVM.

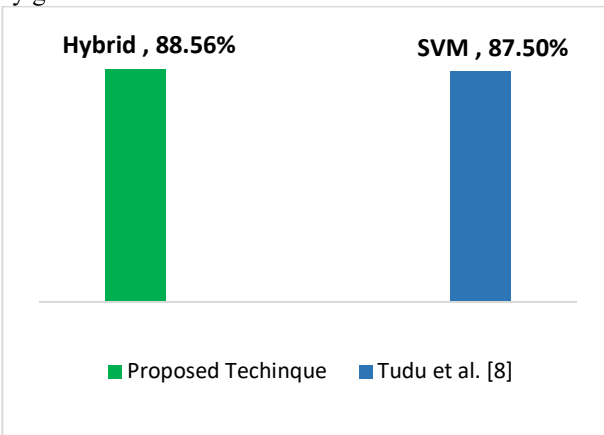

Fig. 7. The best accuracy comparison between the proposed technique and Tudu et al. [8]

Compared with Tudu et al. [8], the word embedding technique we used in our deep learning model helps to get better accuracy than their works on the same dataset. For the other model, we have got 86.14% for CNN, 83.95% for ANN and 86.04% for LSTM based model for 10 categories. They got the lowest accuracy (75.16%) from MNB.

It is becoming increasingly difficult to categorize uncategorized Bangla text data on the web. Machine-learning classifiers are ineffective when it comes to categorizing large datasets. As a counterpoint, deep learning models have the potential to perform better than traditional models. No matter how much data we feed these models, they'll learn well and produce perfect results.

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D. Performance Analysis:

In this sub-section, we have measured the performance of these deep learning based classification models. Precision, recall, and the f1-score are some of the metrics used to evaluate the performance of these models. The equations for these metrics are presented in 1, 2 and 3.

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

Here, TP, FP, TN, FN are respectively true positive, false positive, true negative and false negative. In Table III, the computed average precision, recall and F1-score are shown for these deep learning based classification models.

TABLE III: AVERAGE PRECISION, RECALL, F1-SCORE OF THE DEEP LEARNING MODELS

Models	Precision	Recall	F1-score
Hybrid	0.85	0.85	0.85
CNN	0.83	0.83	0.83
ANN	0.79	0.80	0.80
LSTM	0.81	0.81	0.81

V. CONCLUSION

An effective method for categorizing Bangla text has become necessary in the current period and the main goal of this research work is to develop such a model for categorizing Bangla text data. And we were able to achieve this with excellent accuracy by utilizing deep learning technologies. In this work, we have used the public Bangla dataset to classify into 12 categories where the dataset is not standardized. This dataset contains the same text data into different categories. As a result, models are more prone to misclassification when dealing with this type of text input. That is why models were unable to learn effectively and were misclassified. Among all models, the proposed hybrid model performed best than other single deep learning models and this hybrid model received an accuracy of 84.94 percent for 12 categories and 88.56 percent for 10 categories. Due to the shortage of good resources as well as a standard balanced dataset in the Bangla language, text classification is a big challenge task. These deep learning models can provide greater results than this result if enough resources are available in the Bangla language. As a result, future studies might focus on creating a standard and balanced Bangla dataset, as well as developing a more efficient model for this text categorization problem in NLP.

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